

SciCalc Next CI/CD Report

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1 Introduction

This report documents the implementation of the SciCalc Next scientific calculator application and its accompanying DevOps pipeline. It summarises the project scope, tooling, and the automation built to support repeatable delivery.

2 Setup and Architecture

- Frontend built with Next.js 15 (App Router) and TypeScript.
- Domain logic implemented in a dedicated `src/lib` module for reuse across API routes and unit tests.
- Client UI delivered through a revitalised calculator workspace with expression parsing, quick-operation toggles, and clipboard support.
- Visual styling leverages a glassmorphic theme, responsive panel/sidebar layout, and semantic labelling for improved accessibility.

3 CI/CD Implementation

- Declarative Jenkins pipeline orchestrates installation, linting, testing, and creation of the production build.
- The Docker stage authenticates with Docker Hub using managed credentials before building and pushing `sci-calc-next:latest`.
- A dedicated **Ansible Deploy** stage provisions an ephemeral virtual environment, installs Galaxy collections, and runs the deploy playbook with a vault-sourced password file.

4 Docker and Ansible Deployment

- Multi-stage Dockerfile emits a slim runtime image rooted in `node:20-alpine`.
- The Ansible tree now includes `ansible.cfg`, an inventory, shared group variables, and targeted playbooks in `playbooks/`.
- `playbooks/ping.yml` validates host reachability before any container changes are attempted.
- `playbooks/deploy.yml` logs into Docker Hub when credentials are present, pulls the selected image, and ensures the container binds port 3000 with `'NODE_ENV=production'`.

5 Results and Screenshots

The redesigned interface surfaces contextual guidance, responsive layouts, and clipboard-enabled results. Captured UI previews are tracked under `../screenshots/` for auditability:

- Screenshot 2025-09-30 at 21.03.48 — dark theme overview of the calculator shell.
- Screenshot 2025-09-30 at 21.10.27 — expression parsing and quick action controls in use.
- Screenshot 2025-09-30 at 21.11.19 — factorial input validation state.
- Screenshot 2025-09-30 at 21.50.52 — result panel with copy feedback messaging.

6 Learnings and Improvements

Expression parsing introduced edge cases around negative factorials and whitespace that are now guarded inside the UI before API submission. Future iterations should add automated UI tests to cover copy-to-clipboard fallbacks and broaden accessibility audits.